

Congressional Trade Tracker – Design Doc

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1. Overview

A personal system that monitors public stock-trade disclosures from members of Congress, emails alerts when people we follow trade, and measures how those trades perform after they become public. The analytics answer three questions: which trades are worth copying, when to sell, and whom to follow. Results feed an agent layer that helps refine the strategy over time. Trades are executed **manually in a Fidelity Roth IRA** (Fidelity has no retail trading API). Trading inside the Roth avoids short-term capital gains tax on each trade. The system recommends; a human decides and executes.

Goals

- Get alerted within ~30 minutes of a new House or Senate filing for followed members.
- Measure every disclosed trade against the S&P 500 (SPY) from the date it became public.
- Quantify short-term price inflation caused by copy-traders at the open after disclosure.
- Find exit rules that work best out-of-sample.
- Rank members by how profitable it is to copy them *after* the disclosure lag.
- Expose all of this to agents that summarize, research, and propose strategy changes.

Non-goals (for now)

- Automatic order execution. Humans place all trades.
- Short selling, margin, or options trading (not available or limited in a Roth IRA).
- Reselling or publishing the data (disclosure law restricts commercial use).

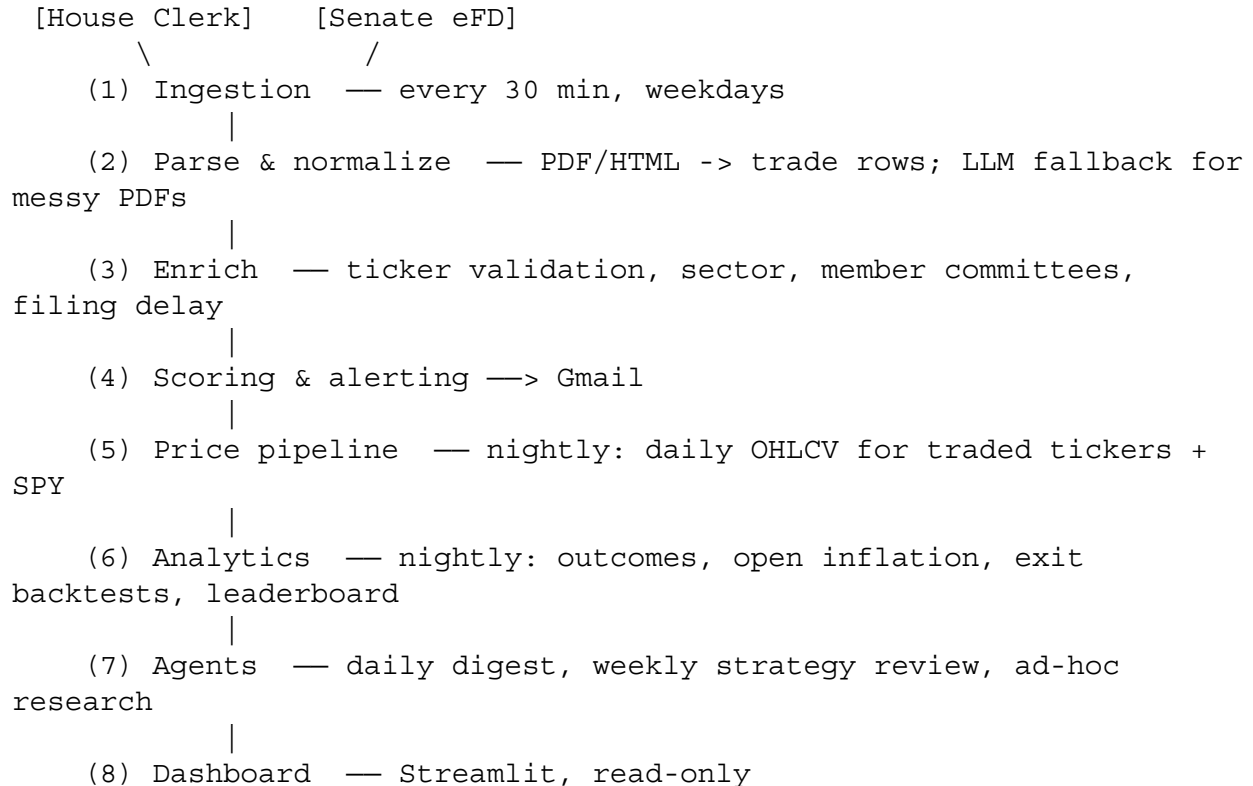
2. Core features

#	Feature	What it does	Key output
F1	Trade alerts	Detects new House/Senate filings, parses trades, filters to the watchlist, and emails a summary with a signal score and suggested exit.	Email alert
F2	S&P benchmark	For each trade,	trade_outcomes table

#	Feature	What it does	Key output
	tracking	computes returns from the disclosure date at fixed horizons and subtracts SPY's return over the same window (abnormal return).	
F3	Open-price inflation	Compares the price at the first open after disclosure with the opens 1, 2, 3 and 5 days later to detect a copy-trader pop.	Best entry delay per member / ticker size
F4	Exit timing	Backtests exit rules (fixed holds, stop-loss/take-profit, trailing stops, ATR stops, exit-on-member-sale) with walk-forward validation.	Recommended exit rule + confidence
F5	Trade scorecard	Labels each disclosed trade (and each of our real trades) as win/loss vs SPY and tracks it over time.	Win/loss history
F6	Member leaderboard	Ranks members by post-disclosure abnormal return, hit rate, sample size, and consistency.	Watchlist recommendations
F7	Agent layer	Agents read the metrics, research new signals, write a weekly strategy review, and propose watchlist / rule changes for human approval.	Daily digest, weekly review

3. Architecture

The system is a set of small, independent pipelines that share one database. Each stage reads from the previous stage's tables and writes its own, so any stage can be re-run or replaced without touching the others.



4. Pipelines

4.1 Ingestion

- **House:** poll the Clerk's yearly filing index (ZIP/XML) for Periodic Transaction Reports (FilingType = P); download each new PTR PDF. If the index turns out to lag the live search page, switch to polling the search page.
- **Senate:** use a requests session that accepts the eFD terms agreement (CSRF token), search for Periodic Transaction Reports by date, and fetch each report's HTML table.
- Store raw files (PDF/HTML) on disk keyed by doc ID so parsing can be re-run without re-downloading.
- Record `first_seen_at` for every filing to measure our detection latency.
- Be polite: polling no more often than every 30 minutes, a descriptive User-Agent, and backoff on errors.

4.2 Parse & normalize

- Electronic House PTRs: pdfplumber text extraction + regex (already prototyped in

- congress_alerts.py).
- Senate electronic reports: BeautifulSoup table parsing.
- Scanned/paper filings: flag as needs_review; phase 2 adds an LLM vision fallback (Claude) to extract rows, with the output marked lower-confidence.
- Normalize: action (BUY / SELL / SELL_PARTIAL / EXCHANGE), owner (self / spouse / joint / dependent), asset type (stock / option / other), amount range as min/max integers.

4.3 Enrich

- Validate tickers against a symbol list; resolve renamed or delisted symbols.
- Add sector/industry, market cap bucket, and the member's party, chamber, and committee assignments.
- Compute filing_delay_days and a committee_relevant flag (e.g., Armed Services member buying a defense stock).

4.4 Scoring & alerting

- v1 score is rule-based: watchlist member, purchase (not sale), larger amount range, shorter filing delay, stock (not fund).
- v2 score comes from the leaderboard and outcome models (F2, F6).
- Email includes: member, ticker, action, amount, trade date, disclosure date, delay, score, suggested entry timing (from F3), suggested exit rule (from F4), and a link to the source filing.
- Position-tracking emails: when we log a purchase, the system watches exit rules daily and emails when one triggers.
- Roth guard: track settled cash (T+1) so suggestions never cause a good faith violation.

4.5 Price pipeline

- Nightly: fetch daily OHLCV for every ticker that appears in a trade, plus SPY, going back far enough to cover all horizons.
- Store adjusted closes for returns and raw opens for entry-price analysis.
- Source: yfinance to start; move to a reliable provider (Alpaca, Polygon, or Tiingo) before trusting results. Delisted tickers are a known gap in free data and cause survivorship bias.

4.6 Analytics

Definitions

- **D0** = first trading-day open after we first saw the filing (the earliest realistic entry).
- **Return(h)** = price at D0 + h trading days / D0 open - 1, for $h \in \{1, 5, 10, 20, 60\}$.
- **Abnormal return(h)** = Return(h) - SPY return over the same window.
- **Win** = abnormal return > 0 at the evaluated horizon (or under the chosen exit rule).
- **Open inflation(k)** = Open(D0) / Open(D0 + k) - 1. Positive means buying k days later was cheaper.
- **Trade-date vs disclosure-date return**: computed for context only (shows how much edge the lag destroys); never used as a tradeable signal.

Jobs

- **Outcomes (F2, F5)**: fill in horizons as they mature; mark trades complete after 60 trading

days.

- **Open inflation (F3):** aggregate by member, market cap bucket, and media attention (e.g., Pelosi vs everyone else).
- **Exit backtests (F4):** grid over rule parameters; walk-forward (train on rolling 2 years, test on the next quarter); report out-of-sample results only; include costs, slippage, and the Roth cash-settlement constraint.
- **Leaderboard (F6):** per member, mean and median abnormal return, hit rate, number of trades, and a shrinkage-adjusted score (empirical Bayes) so members with 3 lucky trades don't top the list. Require a minimum sample size to appear.

4.7 Agent layer

Agents use the Claude API with tool use. They get **read-only** SQL access to the database through a small set of defined tools, plus web search for research. They never place trades and never change rules directly; they write proposals that a human approves.

Agent	Trigger	Job
Filing parser	Scanned/unparseable filing	Extract trade rows from the PDF image; output marked lower-confidence.
Signal researcher	New high-score alert	Pull recent news, earnings dates, and committee context for the ticker; append a short brief to the alert email.
Daily digest	Weekdays, after close	Summarize new filings, open positions, triggered exits, and notable outcomes.
Strategy analyst	Weekly	Review leaderboard and backtest changes; propose watchlist additions/removals and exit-rule changes with evidence.
Journal reviewer	Monthly	Compare our real trades against the system's recommendations and the SPY buy-and-hold baseline.

All agent runs are logged (prompt, tools called, output) in agent_runs for auditing.

5. Data model

Table	Key columns
members	member_id, name, chamber, party, state,

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	committees, active
filings	doc_id, member_id, chamber, filing_date, source_url, raw_path, first_seen_at, parse_status
trades	trade_id, doc_id, member_id, ticker, asset_type, action, owner, tx_date, disclosure_date, amount_min, amount_max, filing_delay_days, committee_relevant, confidence
prices	ticker, date, open, high, low, close, adj_close, volume
trade_outcomes	trade_id, d0_date, d0_open, ret_1/5/10/20/60, abn_ret_1/5/10/20/60, open_infl_1/2/3/5, complete
exit_backtests	run_id, rule, params, train_window, test_window, mean_abn_ret, hit_rate, max_drawdown, n_trades
member_scores	member_id, as_of, n_trades, mean_abn_ret, hit_rate, shrunk_score, rank
watchlist	member_id, added_at, reason, active
alerts	alert_id, trade_id, sent_at, score, suggested_entry, suggested_exit
my_positions	position_id, trade_id (source signal), ticker, buy_date, buy_price, shares, exit_rule, sell_date, sell_price, status
agent_runs	run_id, agent, started_at, inputs, tools_called, output, approved

6. Tech stack

Layer	Choice	Why
Language	Python 3.12	Best ecosystem for scraping, data, and LLM SDKs.
Scraping	httpx/requests, pdfplumber, BeautifulSoup	Simple, reliable for these sources.

Layer	Choice	Why
Storage	SQLite (phase 1) → Postgres on Supabase or Neon free tier (phase 2)	Zero setup locally; hosted Postgres once jobs run in the cloud.
Analytics	pandas + DuckDB; vectorbt optional for exit grids	Fast columnar queries on price history; transparent custom backtests.
Prices	yfinance → Alpaca / Polygon / Tiingo	Free to start; paid data before trusting results.
Scheduling	Local cron (phase 1) → GitHub Actions cron (phase 2)	Free; runs without your computer on.
Alerts	Gmail SMTP with an app password	Already in use; no extra service.
Agents	Anthropic Python SDK (Claude) with tool use	Read-only DB tools + web search; structured outputs for proposals.
Dashboard	Streamlit	Leaderboard, outcomes, and backtest charts in pure Python.
Quality	pytest with saved sample filings; logging to file	Catches parser breakage when source formats change.
Secrets	.env locally; GitHub Actions secrets in the cloud	Never commit passwords or API keys.

Repository layout

```

congress-tracker/
  ingest/          house.py, senate.py
  parse/           house_pdf.py, senate_html.py, llm_fallback.py
  enrich/          tickers.py, committees.py
  prices/          fetch.py
  analytics/       outcomes.py, open_inflation.py, exits.py,
leaderboard.py
  alerts/          score.py, email.py, positions.py
  agents/          tools.py, digest.py, researcher.py, strategist.py
  dashboard/       app.py
  db/              schema.sql, migrations/
  tests/           fixtures/, test_parsers.py
  .github/workflows/ poll.yml (30 min), nightly.yml, weekly.yml

```

7. Rollout plan

Phase	Scope	Done when
1. Alerts	House + Senate ingestion, parsing, email alerts, local cron	Two weeks of alerts with no missed filings vs a public tracker
2. History & prices	Backfill 2020–present filings; price pipeline	Full trade history with prices in DB
3. Analytics	F2, F3, F5, F6; Streamlit dashboard	Leaderboard and inflation results reviewed
4. Exit timing	F4 walk-forward backtests; suggested exits in alerts	Out-of-sample results beat simple fixed-hold baseline, or we learn they don't
5. Agents	Digest, researcher, strategist, journal reviewer	A month of useful weekly reviews
6. Live (small)	Manual trading in a small Roth slice; position tracking	3–6 months compared against SPY buy-and-hold

8. Risks and mitigations

- **Disclosure lag erases the edge.** All metrics are measured from the disclosure date; the go/no-go decision is based on these, not headline returns.
- **Overfitting exit rules and leaderboards.** Walk-forward testing, shrinkage scores, minimum sample sizes, and a small parameter grid.
- **Data quality.** Scanned filings, typos in tickers, amount ranges instead of exact values. Confidence flags, manual review queue, and tests on saved fixtures.
- **Survivorship bias.** Free price data drops delisted tickers; switch to a provider with delisted history before final conclusions.
- **Crowding.** Popular members' trades may pop at the open; F3 tells us whether to delay entry.
- **Roth constraints.** Cash-account settlement (good faith violations), no shorting, limited contribution room. Settled-cash tracking and a small allocation.
- **Source changes / rate limits.** Polite polling, raw-file caching, parser tests, alert on repeated failures.
- **Legal use.** Personal research only; no resale or publication of the data.
- **Agent errors.** Read-only tools, human approval for every change, full run logging.

9. Open questions

- Initial watchlist: follow everyone and let F6 decide, or start with a hand-picked list?

- How much Roth money to allocate in phase 6, and per-position size limits?
- Include options trades as signals (translated to buying the underlying stock)?
- Treat sales as exit signals for our positions, or as bearish signals to avoid a ticker?
- Budget for paid price data and hosted database once past phase 2?